

DRILLING INCIDENT REPORT - STUCK PIPE

SYNTHETIC DEMONSTRATION DOCUMENT - NOT REAL OPERATIONAL DATA (SIH PROTOTYPE)

WELL IDENTIFIER: NWIS-W004

FORMATION: Demo-Kopili Shale

EVENT TYPE: STUCK_PIPE

OPERATIONAL DATE: 2021-09-12

MEASURED DEPTH: 3520.0 m

SEVERITY LEVEL: HIGH

INCIDENT DESCRIPTION

While pulling out of hole (POOH) for bit change at 3520m in Demo-Kopili Shale, drillstring became mechanically stuck. Overpull exceeded 75 tons with zero upward movement. Surface rotary torque spiked up to 39.5 kNm prior to string stalling.

ROOT CAUSE INVESTIGATION

Highly reactive smectite-rich Kopili marine shale experienced severe chemical hydration and borehole breakout due to insufficient potassium chloride (KCl) inhibition in the mud. Large cavings sloughed into the borehole cavity, causing severe mechanical pack-off around the 8-1/2 inch BHA.

REMEDIAL MITIGATION

1. Pumped 40 bbls glycol-weighted lubricant pill across the stuck interval.
2. Engaged hydraulic drilling jar upward at 70 tons combined with string torque working.
3. String came free after 6.5 hours of continuous jarring operations.
4. Circulated high-viscosity sweep to carry shale cavings to surface; densified mud to 1.35 SG.